

SHAHID ALI

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Pakistan Engineering Council (PEC) Reg. No.: METAL/04064

Materials Scientist

Polymer electrolytes | Multiscale modeling (DFT→MD→COMSOL) | Electrochemical characterization | Structural metallurgy

EDUCATION

National University of Science & Technology (NUST MISIS)

Sept 2024 – Aug 2026

MSc in Advanced Materials Science (CGPA: 4.58/5.0 — Excellent)

Moscow, Russia

- **Thesis:** Fabrication, Optimization, and Theoretical Structure Studies of PVA-Based Electrospun Fibrous Membranes for Ion Conduction Applications | Advisor: Prof. Alexey I. Salimon
- **Coursework:** Materials Modeling & Simulation, Surface & Interface Science, Diffusion in Solids, Modern Structural Characterization, Thermodynamics

Mehran University of Engineering & Technology (MUET)

Jan 2011 – Dec 2015

B.E. in Metallurgy & Materials Engineering (First Division)

Jamshoro, Pakistan

- **Thesis:** Enhancement of Mechanical Properties of TIG-Welded Aluminium 6061 via Heat Treatment Optimization

Sindh Board of Technical Education

2012 – 2014

Diploma of Associate Engineer, Mechanical Technology (First Division, 67.3%)

Karachi, Pakistan

University of Sindh

2011 – 2013

B.Com (Bachelor of Commerce)

Jamshoro, Pakistan

RESEARCH EXPERIENCE

Graduate Research Assistant — Department of Physical Chemistry

Sept 2024 – Present

National University of Science and Technology MISIS

Moscow, Russia

- **Multiscale materials modeling:** Executed 31 DFT calculations (ORCA 6.1.1: B3LYP-D3BJ, B97X-D3, def2-TZVPP; NBO, ELF, NEB transition states) on PVA/HPO systems. Validated with 500 ps LAMMPS MD simulations (NPT, OPLS-AA; P-O RDF 4.25 Å, MSD <3 Å²) and COMSOL fuel-cell modeling (0.7% deviation vs. EIS).
- **Membrane fabrication & characterization:** Electrospun 50+ PVA/citric acid nanofiber membranes (150–400 nm fibers); optimized voltage, thermal crosslinking (120–160 °C), HPO doping. Characterized via SEM (1000+ images), XRD (Rietveld), FTIR-ATR, EIS (41.4 mS/cm), TGA, mechanical testing (0.5–2.0 GPa).
- **Data automation:** Developed Python workflows (NumPy, Pandas, Matplotlib, SciPy) for automated EIS fitting and SEM analysis, reducing processing time by 70%.

Undergraduate Research Assistant — Department of Metallurgy & Materials

Jan 2014 – Dec 2015

Mehran University of Engineering & Technology (MUET)

Jamshoro, Pakistan

- **Aluminium alloy optimization:** Investigated post-weld heat treatment (150–200 °C, 2–8 h) effects on Al6061 microstructure and mechanical properties via metallography, hardness mapping (Vickers), tensile testing.
- **Machine learning:** Developed Bayesian optimization model (Gaussian processes) to predict microhardness from aging parameters; achieved 15% error reduction vs. empirical models.
- Characterized failure modes via SEM/EDS; correlated precipitate morphology (Mg₂Si) with hardness; identified optimal aging (175 °C, 6 h, 20% hardness increase).

Research Intern — Center for Digital Engineering

Feb 2025 – Aug 2025

Skolkovo Institute of Science and Technology (Skoltech)

Moscow, Russia

- Completed: *Technical Analysis of Actuator Solutions for Ultrasonic Signal Excitation in Railway Infrastructure*. Calibrated 8+ piezoelectric Langevin transducers (95% signal accuracy); integrated into SHM frameworks.
- Performed MATLAB/Python signal processing (FFT, wavelet); developed ML pattern recognition (SVM, random forest) reducing false positives by 35%.

PUBLICATIONS

Under Review: “Covalent Phosphorylation of PVA Eliminates Acid Leaching without Reducing Proton Donor Activity.” *J. Molecular Modeling*.

Under Review: “Molecular-Level DFT Analysis of Phosphoric Acid Binding and Grotthuss Proton Transfer in PVA Membranes.” *Polymer Bulletin*.

Under Review: “Facile Fabrication of Electrospun PVA/Citric Acid Membranes: Morphological Evolution via Optimized Doping.” *J. Membrane Science & Research*.

Decision Pending: “A Novel Approach to Microhardness Prediction in Al6061 Welds: Synergistic Use of Synthetic Data, Experimental Validation, and Bayesian Optimization.” *Intl. J. Lightweight Materials*.

In Preparation: “Multiscale Validation of Covalent Phosphate Immobilization: DFT to MD to COMSOL.” Target: *ACS Appl. Mater. Interfaces*.

TECHNICAL SKILLS

Computational: DFT (ORCA: B3LYP-D3BJ, B97X-D3, M06-2X; NBO, ELF, NEB), MD (LAMMPS: OPLS-AA, NPT/NVT, RDF, MSD), COMSOL (Nernst–Planck, Butler–Volmer), Python (NumPy, Pandas, Matplotlib, SciPy, scikit-learn), MATLAB, R, Tableau, Abaqus, AutoCAD, LaTeX, Linux/SLURM

Characterization: SEM/EDS, TEM, XRD (Rietveld), FTIR-ATR, optical microscopy, EIS (Nyquist/Bode, equivalent circuits), CV, TGA/DSC, contact angle

Mechanical Testing: Tensile (UTM), Vickers microhardness, nanoindentation, fatigue testing

Fabrication & NDT: Electrospinning, thermal crosslinking, metallography, heat treatment, ultrasonic testing, magnetic particle inspection, dye penetration

PROFESSIONAL EXPERIENCE

Ministry of Defence, Government of Pakistan

Mar 2021 – Aug 2021

Signal Analyst (BPS-11)

Rawalpindi/Islamabad, Pakistan

- Monitored/analyzed signal data for defense communication; developed automated dashboards for real-time decision support.
- Improved workflows via Python scripts (25% time reduction); applied signal processing skills in Skoltech SHM project.

Sindh Education & Literacy Department

2021 – Present (On Leave)

Project Monitoring & Evaluation Assistant

Dadu, Pakistan

- Digitized 50+ project records (40% reporting time reduction); implemented data-driven monitoring (15% completion rate increase across 15 districts).

Ibrahim Hussain Al Qureshi Est. (Al Qureshi Fruits)

Jun 2016 – Jan 2018

Account Officer

Mecca, Saudi Arabia

- Managed financial transactions, A/R, A/P, bank reconciliations; prepared monthly statements; ensured Saudi regulatory compliance.
- Gained cross-cultural KSA work experience; Arabic language exposure; understanding of Middle Eastern business practices.

Aftab Engineering Pvt. Ltd.

2014 – 2015

Quality Assurance Intern

Karachi, Pakistan

- Inspected 1000+ components (20% QC efficiency improvement); optimized processes (15% waste reduction in machining).

ENTREPRENEURSHIP & PROJECTS

Off-Grid Solar Energy Startup (Co-founder)

2018 – 2020

Technical Lead & Project Manager

Dadu, Pakistan

- Deployed 15+ solar systems for 200+ rural residents; managed \$20K budget, 5-person team; designed 24-hour battery autonomy (90% satisfaction).

Injection Moulding Machine — Future Engineers' Expo

2014

Project Builder

MUET, Pakistan

- Constructed working injection moulding machine for university event; produced promotional keychains on-site; demonstrated polymer processing principles.

PROFESSIONAL CERTIFICATIONS & ONLINE COURSES

- **2025:** Principles of Chemical Science (MIT, USA)
- **2025:** Chemistry by Khan Academy (CURSA)
- **2025:** Renewable Energy Engineering (NPTEL, India)
- **2025:** 3D Printing (HP LIFE)
- **2025:** AI & Data Science (6-month, Codanics)
- **2024:** R Language Course (Codanics)
- **2015:** AutoCAD (MUET)
- **2014:** Magnetic Particle Inspection & Dye Penetration (MMAC, MUET)
- **2014:** Numerical Methods & Computation (MUET)

SCHOLARSHIPS & AWARDS

- **2024:** Open Door Scholarship (Top 5%, Russia)
- **2023:** ISCO Chemical Eng. Scholarship (Iran)
- **2023:** IAC Nanomaterials Scholarship (Iran)
- **2023:** Early Bird Scholarship (LUT, Finland)
- **2023:** Oulu Mining Eng. Scholarship (Finland)
- **2016:** UNESCO/Poland Fellowship (1 of 20 globally)
- **2016:** PM Youth Program (Pakistan)
- **2015:** Panaflex Shield of Merit (MUET)
- **2014:** Cricket Tournament Cup (MUET)
- **2007:** Gold Medal, Basketball (All Sindh, Pakistan)

CONFERENCES & EXTRACURRICULAR ENGAGEMENTS

- **2025:** Water Policy Development (UN SDG-6, A-LAB, NUST MISIS) — Researcher
- **2025:** Skoltech SMILES Summer School
- **2024:** Skoltech Frontier of Progress
- **2024:** Matter & Materials Symposium
- **2024:** Intl. Materials Science Conference
- **2014:** Volunteer — Career Fair, Festivals (MUET)
- **2013:** Organizer — 20+ study tours, competitions (MMAC, MUET)

Fluent: English, Urdu/Hindi, Sindhi | Intermediate: Russian | Basic: Arabic, Persian | References available upon request.